Deductive Interpolation from Horn Semantics

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We will report on work in progress, in which we consider semantics of a logic in a class of first order structures axiomatized by implications between atomic formulas, a *Horn class*. We give conditions on such a semantics which ensure that an amalgamation property for the Horn class implies Craig interpolation for the logic.

This generalizes - from algebraic structures to more general first order structures - the well-known result that for an algebraizable logic the amalgamation property for the associated class of algebras implies the deductive interpolation property.

Horn Semantics

Let L be a logic over a signature Σ. Recall that an algebraic semantics for a logic L is a translation from formulas of L to sets of equations over the signature of L (i.e. atomic formulas of the first order language associated to Σ), commuting with substitution, and such that inference in the logic corresponds, under this translation, to inference in the equational logic in a quasivariety $\mathbf{K}$. It is an equivalent algebraic semantics it induces an isomorphism of theory lattices.

For a Horn semantics one considers an expansion Σ^+ of Σ by relation symbols. One replaces the quasivariety $\mathbf{K}$ by a class of first order structures axiomatized by rules of the form $\bigwedge_{i\in I} P_i(\vec{x}) \vdash P(\vec{x})$, with P_i, P atomic formulas over Σ^+, not equivalent to $\perp$. Such a class is called a *Horn class*. Now one translates formulas of L to sets of atomic formulas over Σ^+. An equivalent Horn semantics is, again, one inducing an isomorphism of theory lattices.

Horn Semantics via Filter Pairs

The situation of an algebraic semantics has been abstracted into the notion of *filter pair* in [**AMP1**], [**AMP2**], [**AMP3**]: A filter pair is a functor $G\colon \Sigma\text{-Str} \to \text{AlgLat}$, from Σ-structures to algebraic lattices, together with a natural transformation to the power set functor $i\colon G \to \wp$, which preserves infima and κ-directed suprema, objectwise. To capture algebraic semantics, one takes the functor that takes an algebra A to the lattice of congruences relative to $\mathbf{K}$: $G := Co_{\mathbf{K}}(A) := \{\theta \mid A/\theta \in \mathbf{K}\}$. Then an equivalent algebraic semantics is one where the natural transformation i is objectwise injective.

Horn Semantics arises by replacing the congruence lattice with a lattice of atomic theories over Σ^+: For a Σ^+-structure A let Σ_A be the signature Σ enhanced by a constant symbol for each element of A, and let $\mathbb{T}_A$ be the collection of all atomic sentences true in A. Consider a set $\mathbb{T}$ of rules defining a Horn class. We then define

$$G^{\mathbb{T}}(A) := \{\mathbb{T}\text{-closed sets of atomic sentences over } \Sigma_A \text{ that contain } \mathbb{T}_A\}.$$

Such a filter pair is called *Horn filter pair*. A presentation of a logic by a Horn filter pair is a Horn semantics for it, and it is an equivalent Horn semantics, if the natural transformation i is injective.

Examples

- For an algebraic signature $\Sigma^+ = \Sigma$ a Horn semantics is precisely an algebraic semantics, and an equivalent Horn semantics is an equivalent algebraic semantics.
- For $\Sigma^+ = \Sigma \cup \{F\}$ an expansion of the signature with a unary relation symbol one can define an equivalent Horn semantics corresponding to matrix semantics.
- For $\Sigma^+ = \Sigma \cup \{\leq\}$ an expansion of the signature with an inequality symbol, and Horn rules demanding that this be an order relation, a Horn semantics is precisely an order algebraic semantics, and an equivalent Horn semantics corresponds precisely to an order algebraizable logic in the sense of [**Raf**]

Amalgamation and Interpolation

Using the formalism of filter pairs, we can prove a general interpolation result:
We shall say that a class $\mathbf{K}$ of Σ^+-structures has the *atomic amalgamation property* if, given $A, B, C \in \mathbf{K}$ and maps $i_B \colon A \to B$, $i_C \colon A \to C$ that preserve and reflect the validity of atomic formulas (atomic embeddings), there exist a Σ-structure $D \in \mathbf{K}$ and atomic embeddings $e_B : B \to D$, $e_C : C \to D$ such that $e_B \circ i_B = e_C \circ i_C$.

Theorem Let $(G^{\mathbb{T}}, i)$ be a Horn semantics for a logic L. Suppose that the filter pair $(G^{\mathbb{T}}, i)$ has the "theory lifting property". If $\mathbf{K} := \mathrm{Mod}(\mathbb{T})$ has the atomic amalgamation property, then the logic L associated to $(G^{\mathbb{T}}, i)$ has the deductive interpolation property.

The "theory lifting property" is a technical condition, satisfied by every filter pair presenting an equivalent Horn Semantics, but also in some other cases.

Examples The above theorem specializes to the following statements:

- the well-known statement that for algebraizable logics, the amalgamation property entails deductive interpolation
- the well-known statement that the theory amalgamation property entails deductive interpolation
- a corresponding statement for order algebraizable logics
- the statement that for logics with an algebraic semantics in a regular variety, the amalgamation property entails deductive interpolation

In the talk we will introduce the setup of filter pairs and explain the above results and examples.

References

[**AMP1**] Arndt, P., Mariano H., Pinto D., Finitary Filter Pairs and propositional logics, South American Journal of Logic, vol. 4, Nr. 2 p. 257-280 (2018)

[**AMP2**] Arndt, P., Mariano H., Pinto D., Filter Pairs and natural extensions of logics, Archive for Mathematical Logic volume 62, pages 113-145 (2023)

[**AMP3**] Arndt, P., Mariano H., Pinto D., Congruence Filter pairs, equational filter pairs and adjoints, Journal of Logic and Computation, vol. 34 (2), (2024)

[**Raf**] Raftery, J.G., Order algebraizable logics, Annals of Pure and Applied Logic 164 (2013) 251-283